Xuyuan Liu

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Research Interests

My research focuses on **the science of Language Models**, with an emphasis on understanding their behavior during both training and inference by probing their internal mechanisms. My goal is to **understand how LLMs work internally** and to translate that understanding into technical methods that enhance their reasoning, reliability, and personalization capabilities.

Education

- **Dartmouth College** — *Sept. 2023 – Present*
  Ph.D. in Computer Science — Hanover, NH
  ◦ Advisor: Prof. Yujun Yan
- **Nankai University** — *Sept. 2019 – June 2023*
  B.Eng. in Computer Science — Tianjin, China

Internship Experience

- **Amazon** — *June 2026 – Sep. 2026*
  Applied Scientist Intern — Seattle, WA
  - Developing an agentic training system that tackles the *credit assignment* problem in long-horizon tool-use training, attributing sparse trajectory-level rewards to individual tool-invocation steps for more reliable reward attribution.
- **NEC Laboratories America** — *June 2025 – Sep. 2025*
  Research Scientist Intern — Princeton, NJ
  - Proposed a framework for *lifelong LLM knowledge memory personalization* via hidden-state interventions, enabling precise, localized, and lightweight knowledge updates without low-cost training.
  - **Outcome: [CP.5]** (Accepted to ACL 2026, Oral).

Publications & Preprints

* Equal Contribution; S=In Submission, CP=Conference Proceedings

[CP.6] **Xuyuan Liu**, Xinshuai Dong, Elynn Chen, and Yujun Yan. **When Form Changes but Logic Doesn't: Building Logic-Invariant LLMs through Structures**. In *Proceedings of the Fortieth Annual Conference on Neural Information Processing Systems* (**NeurIPS 2026**).
LLM Reasoning · Symbolic Systems · Multi-Modality Learning

[CP.5] **Xuyuan Liu**, Shengyu Chen, Xinshuai Dong, …, Yujun Yan, Haifeng Chen, Zhengzhang Chen. **Representation Interventions Enable LLM's Lifelong Knowledge Memory Control**. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics* (**ACL 2026**). Oral.
LLM Personalization · LLM Memory Customization · Efficient LLM

[CP.4] **Xuyuan Liu**, Lei Hsiung, Yaoqing Yang, and Yujun Yan. **Spectral Insights into Data-Oblivious Critical Layers in Large Language Models**. In *Findings of the Association for Computational Linguistics* (**ACL 2025**). Also presented at the *UniReps Workshop @ NeurIPS*.
LLM Training Behavior · Efficient LLM

[CP.3] **Xuyuan Liu**, Yinghao Cai, Qihui Yang, and Yujun Yan. **Exploring Consistency in Graph Representations: From Graph Kernels to Graph Neural Networks**. In *Proceedings of the Thirty-Eighth Annual Conference on Neural Information Processing Systems* (**NeurIPS 2024**).
Representation Learning

[CP.2] Jinhang Chai, **Xuyuan Liu**, Elynn Chen, and Yujun Yan. **Low-Rank Plus Sparse Matrix Transfer Learning under Growing Representations and Ambient Dimensions**. In *Submission.*
Representation Learning

[CP.1] Zichen Liu, **Xuyuan Liu**, Yanlong Wen, Guoqing Zhao, Fen Xia, and Xiaojie Yuan. **TreeMAN: Tree-Enhanced Multimodal Attention Network for ICD Coding**. In *Proceedings of the 29th International Conference on Computational Linguistics* (**COLING 2022**). Oral.
Multi-Modality Learning

Honors, Awards, and Grants

- **Oral Presentation.** ACL 2026
- **Silver Reviewer Award.** ICML 2026
- **Guarini Travel Grant (2024).** Guarini School of Graduate and Advanced Studies, Dartmouth College.
- **Dartmouth Fellowship (2023).** Guarini School of Graduate and Advanced Studies, Dartmouth College.
- **Academic Excellence Scholarship.** Nankai University (Top 7/130).

Teaching Experience

Head Teaching Assistant, *Deep Learning* (Spring 2024); Head Teaching Assistant, *Machine Learning & Statistical Data Analysis* (Fall 2024), Dartmouth College.

Professional Services

- **Reviewer:** ICLR 2025–2026, NeurIPS 2024–2025, ICML 2025–2026, KDD Dataset & Benchmark Track, TMLR, TPAMI

Skills

- **Programming Languages:** C, C++, Python, Java, SQL, Shell Scripting
- **ML/LLM Frameworks and Tools:** PyTorch, vLLM, HuggingFace, VeRL, OpenRLHF
- **Tools:** Git, LaTeX, Slurm